

Week 2 - Day 8b Supplementary: More Advanced Images

```
from simpleimage import SimpleImage
```

Exercise 1: Making an Ethiopian Flag []

1.1 Write code that draws the background of the Ethiopian flag.

```
# TODO: Your code here

size = 11
image = SimpleImage.blank(size, size)

# Create the image here

image.show(resize_width=200)
```

1.2 We're missing the blue emblem in the center. Let's figure out how to draw a circle.

If the size of the image is `size`, what are the coordinates of the image center?

1.3 If we have a pixel at (x, y) , how do we compute the squared distance between that pixel and the image center?

1.4 Now draw a circle.

```
# TODO: Your code here

size = 11
image = SimpleImage.blank(size, size)

# Create the image here

image.show(resize_width=200)
```

1.5 Now use your code to add a blue circle to your Ethiopian flag!

```
# TODO: Your code here

size = 11
image = SimpleImage.blank(size, size)
```

```
# Create the image here  
image.show(resize_width=200)
```

1.6 If you want, write code that adds a few pixels for the yellow star to the center of the blue circle!

```
# TODO: Your code here  
size = 11  
image = SimpleImage.blank(size, size)  
  
# Create image here  
image.show(resize_width=200)
```

Exercise 2: Making a Liberian Flag []

2.1 Draw horizontal bars, as shown here.

```
# TODO: Your code here  
size = 11  
image = SimpleImage.blank(size, size)  
  
# Create image here  
image.show(resize_width=200)
```

2.2 Draw the flag of Liberia, as shown here.

```
# TODO: Your code here  
size = 11  
image = SimpleImage.blank(size, size)  
  
# Create image here  
image.show(resize_width=200)
```

Exercise 3: Gradients []

3.1 Remember the gradients from 7a? Now, we want make a two-directional gradient using cyan and red.

```
# TODO: Your code here
size = 11
image = SimpleImage.blank(size, size)

# Create image here

image.show(resize_width=200)
```

3.2 Draw the horizontal bars with a smooth color gradient?

```
# TODO: Your code here
size = 11
image = SimpleImage.blank(size, size)

# Create image here

image.show(resize_width=200)
```

3.3 Draw the bars while cycling through the colors in the list `cycle`.

```
# TODO: Your code here
colors = [
    [255, 0, 0],
    [255, 255, 0],
    [0, 255, 0],
    [0, 255, 255],
    [0, 0, 255],
    [255, 0, 255],
]
size = 11
image = SimpleImage.blank(size, size)

# Create image here

image.show(resize_width=200)
```

Exercise 4: Spirals [] [] [] []

4.1 Draw a snake, as shown below.

```
# TODO: Your code here
size = 11
image = SimpleImage.blank(size, size)
```

```
# Create image here  
image.show(resize_width=200)
```

4.2 Draw a spiral, as shown below.

```
# TODO: Your code here  
size = 11  
image = SimpleImage.blank(size, size)  
  
# Draw spiral here  
image.show(resize_width=200)
```

4.3 Modify your code so the spiral uses a color gradient, as shown below.

```
# TODO: Your code here  
size = 11  
image = SimpleImage.blank(size, size)  
  
# Create image here  
image.show(resize_width=200)
```

4.4 Draw another spiral over the previous spiral with another color gradient, as shown below.

```
# TODO: Your code here  
size = 11  
image = SimpleImage.blank(size, size)  
  
# Create image here  
image.show(resize_width=200)
```

4.5 Draw a snake that fills every line with a color gradient from black to white.

```
# TODO: Your code here  
size = 11  
image = SimpleImage.blank(size, size)  
  
# Create image here
```

```
image.show(resize_width=200)
```

Exercise 5: Interpolate [] []

5.1 Define a function `interpolate_linear` that takes two colors and a value `t` and computes a new color that is between `color1` and `color2`, with the color components being $\text{color2}[i]*t + \text{color1}[i]*(1-t)$.

```
# TODO: Your code here
def interpolate_linear(color1, color2, t):
    pass
```

5.2 Use this function to draw a vertical gradient from magenta to green, as shown below.

```
# TODO: Your code here
size = 11
image = SimpleImage.blank(size, size)

# Create image here

image.show(resize_width=200)
```

5.2 Draw a colored snake that fills every line, as shown below.

Use the color cycle from before, and the function `interpolate_linear` to interpolate between consecutive colors.

```
# TODO: Your code here
cycle = [
    [255, 0, 0],
    [255, 255, 0],
    [0, 255, 0],
    [0, 255, 255],
    [0, 0, 255],
    [255, 0, 255],
]
size = 11
image = SimpleImage.blank(size, size)

# Create image here

image.show(resize_width=200)
```

Congratulations []